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MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B. TECH (ELECTRICAL & ELECTRONICS ENGINEERING)

END SEMESTER EXAMINATIONS NOVEMBER 2024

RENEWABLE ENERGY [ELE 4086]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 21 Nonember 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A Describe solar cells' construction and working principle with a neat diagram and explain its V-I Characteristics. ***(04)***

1B Describe the construction and working principle of the Angstrom Pyrheliometer with a neat figure. ***(03)***

1C Estimate the daily average global radiation on a horizontal surface at Baroda 22^0 N during March 16 if the average sunshine hours per day is 9.5. Assume a=0.28, b=0.48, and given H_0=11911 kJ/m^2 ***(03)***

2A Describe the power generation in a central receiver power plant with a neat figure. ***(03)***

2B Derive an expression for the Buck-Boost converter for solar PV application. ***(04)***

2C. A PV System feeds a DC motor to produce 2 HP power at the soft. Motor efficiency is 80%, and each module has 36 solar cells arranged in a 9 x 4 matric. The cell size is 125 mm x 125 mm & the cell efficiency is 14%. Calculate the no of modules required in the PV array. Assume global incident radiation usually to the panel as 1 kW/m^2 ***(03)***

3A. Derive an expression for the Maximum Power, Torque, and Thrust generation in the windmill with the help of a wind velocity diagram. ***(03)***

3B. Wind at one standard atmospheric pressure and 25^0 c has a velocity of 25m/s calculate: Given turbine diameter=120 m

a. Total power and power density

b. Max power and power density ***(03)***

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c. Total power when the η is 35%. Given: $\dot{\rho}=\frac{P}{RT}$ Where R=0.287 kJ/kg, P=1.01325 X 10^5

3C. Describe the following wind turbine construction and working principles with their figures and list their advantages and disadvantages.

a. Savonius Rotor
b. Darrieus type m/c's ***(04)***

4A Describe the construction and working of the (KVIC) floating drum-type biogas power plant with its neat diagram. List out its advantages and disadvantages. ***(04)***

4B Describe factors to be considered in the site section of a Biogas plant. ***(03)***

4C The following data are given for a family biogas digester suitable for the o/p of 10 cows: The retention time is 25 days, temp is 28^0 c, dry matter consumed per day is 2 kg, and biogas yield is 0.25 m^3 per kg. The burner's efficiency is 60%, the methane proportion is 0.7. The heat of combustion of methane is 38 MJ/m^3. Calculate the amount of energy generated in the digester and also its volume. ***(03)***

5A Describe the power generation of the following Tidal plant with their neat diagram.

a. Double basin linked basin plant

b. Double basin paired basin plant ***(04)***

5B Describe the construction and working principle of the Hybrid Ocean Thermal Energy Conversion(OTEC) power plant that overcame the limitations of closed and open cycle OTEC with a neat figure. ***(03)***

5C Describe the power generation using the Flash steam open systems (liquid dominated) of geothermal energy conversion with a neat diagram and mention its limitations. ***(03)***